

Course

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1.1.1. Calculate Momentum17:15

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula:

$$p = m \times v$$

where:

- m is the mass of the object (in kilograms).
- v is the velocity of the object (in meters per second).

Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Sample Test Cases

calculate...

```
1 mass=float(input())
2 velocity=float(input())
3 momentum=mass*velocity
4 print(f"{momentum:.2f}kgm/s")
```

Average time0.003 s3.00 msMaximum time0.005 s5.00 ms

2 out of 2 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 15 msDebug

Expected output

Actual output

5.0

5.0

10.0

10.0

50.00kgm/s

50.00kgm/s

Test case 22 ms

TerminalTest cases

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1.1.5. Multiplication Table

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

Print the multiplication table for the given number in the format:

`<number> x <multiplier> = <result>`

Note:

The character "x" is used in the output to represent multiplication.

Refer to the visible test-cases for more clarity.

Sample Test Cases

multiplia...

```
1 n = int(input())
2 for i in range(1,11):
3     print(f"{n} x {i} = {n*i}")
```

Average time0.002 s2.25 msMaximum time0.003 s3.00 ms

2 out of 2 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 12.00sDebug

Expected output	Actual output
8 x 1 = 8	8 x 1 = 8
8 x 2 = 16	8 x 2 = 16
8 x 3 = 24	8 x 3 = 24
8 x 4 = 32	8 x 4 = 32
8 x 5 = 40	8 x 5 = 40

TerminalTest cases

Thunderstorm w...In effect

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1.1.4. Reverse a Number

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

The input is a single integer.

Output Format:

The output prints a single integer, which is the reversed number.

Sample Test Cases

reverseN...

```
1 num=int(input().strip())
2 reversed_num = int(str(num)[::-1])
3 print(reversed_num)
```

Average time0.002 s2.20 msMaximum time0.003 s3.00 ms

2 out of 2 shown test case(s) passed3 out of 3 hidden test case(s) passed

Test case 13 msDebug

Expected output

Actual output

-5367-5367

76357635

Test case 22 ms

Terminal

Test cases

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1.1.3. Days between Two Dates15.00

Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format *YYYY – MM – DD*.
- The second line contains the second date in the format *YYYY – MM – DD*.

Output Format:

- An integer representing the number of days between the given dates.

Note:

- The first date should always be considered earlier than the second date

Sample Test Cases

noOfDays...

```
1 from datetime import datetime
2
3 date1=input().strip()
4 date2=input().strip()
5 d1=datetime.strptime(date1,"%Y-%m-%d")
6 d2=datetime.strptime(date2,"%Y-%m-%d")
7 difference=d2-d1
8 print(difference.days)
```

Average time0.006 s6.00 msMaximum time0.016 s16.00 ms

2 out of 2 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 115.00msDebug

Expected output

Actual output

2014-07-022014-07-02

2014-11-022014-11-02

123123

Test case 23.00ms

Terminal

Test cases

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1.1.2. Conditional Calculation Based on the Number of Digits

Write a Python program that accepts an integer n as input. Depending on the number of digits in n .

Constraints:
 $1 \leq n \leq 999$

Input Format:

- The input consists of a single integer n .

Output Format:

- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Sample Test Cases

condition...

```
1 #write your code here
2 import math
3 n = int(input())
4 if 0 <= n <= 999:
5     if 0 <= n <= 9:
6         print(n * n)
7     elif 10 <= n <= 99:
8         print(f"{math.sqrt(n):.2f}")
9     else:
10        print(f"{n ** (1/3):.2f}")
11 else:
12     print("Invalid")
13
```

Average time
0.002 s
1.86 ms

Maximum time
0.003 s
3.00 ms

4 out of 4 shown test case(s) passed
3 out of 3 hidden test case(s) passed

Test case 1 3 ms Debug

Expected output
81

Actual output
81

Test case 2 1 ms

Test case 3 2 ms

Terminal Test cases

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1.2.1. Pass or Fail

Write a Python program that accepts the number of courses and the marks of a student in those courses. The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n , the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses.

passorFa...

```
1 def calculate_grade(num_courses, marks):
2     if len(marks) != num_courses:
3         return "Error: Number of marks does not match number of courses."
4
5     if any(mark < 40 for mark in marks):
6         return "Fail"
7
8     aggregate_percentage = sum(marks)/num_courses
9
10    if aggregate_percentage > 75:
11        grade = "Distinction"
12    elif aggregate_percentage > 60:
13        grade = "First Division"
14    elif aggregate_percentage > 50:
15        grade = "Second Division"
16    elif aggregate_percentage >= 40:
17        grade = "Third Division"
18    else:
19        grade = "Fail"
20
```

Average time

0.003 s

2.70 ms

Maximum time

0.005 s

5.00 ms

5 out of 5 shown test case(s) passed

5 out of 5 hidden test case(s) passed

Test case 1

Expected output

4

55.65 78.89

Aggregate Percentage: 71.75

Grade: First Division

Actual output

4

55.65 78.89

Aggregate Percentage: 71.75

Grade: First Division

Test case 2

Terminal

Test cases

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1.2.2. Fibonacci Series

075%

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

- A single integer n representing the number of terms to generate.

Output Format:

- A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases

fibonacci...

```
1 def fibonacci(n):
2     if n == 1:
3         return 0
4     elif n == 2:
5         return 1
6     else:
7
8         return fibonacci(n - 1) + fibonacci(n - 2)
9
10 n = int(input())
11 for i in range(1, n + 1):
12     print(fibonacci(i), end=" ")
```

Average time0.006 s6.00 msMaximum time0.016 s16.00 ms

2 out of 2 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 12 msDebug

Expected output5Actual output0 1 1 2 3

Test case 22 ms

TerminalTest cases

< PrevResetSubmitNext >

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1.2.3. Pattern - 1075%

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

The input is an integer, representing the number of rows in the pattern.

Output Format

The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases

rightangl...

1#Type Content here...

2n = int(input())

3for i in range(1, n+1):

4for j in range(i):

5print("*",end=" ")

6print()

Average time0.002 s2.17 ms

Maximum time0.003 s3.00 ms

2 out of 2 shown test case(s) passed

4 out of 4 hidden test case(s) passed

Test case 13msDebug

Expected output

Actual output

*
**

*
*
*
*
*

TerminalTest cases

Thunderstorm w...
In effect

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1.2.4. Pattern - 2

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

The input is an integer, representing the number of rows in the pattern.

Output Format:

The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

Refer to the displayed test cases for the sample pattern.

Sample Test Cases

numberP...

```
1 # Type Content here...
2 # Type Content here...
3 def print_triangle_pattern(rows):
4     for i in range(1, rows + 1):
5         # Generate the numbers for the current row
6         for j in range(1, i + 1):
7             print(j, end=' ')
8         # Move to the next line after each row
9         print()
10
11 # Input: number of rows
12 try:
13     num_rows = int(input())
14     if num_rows < 1:
15         print("Please enter a positive integer.")
16     else:
17         print_triangle_pattern(num_rows)
18 except ValueError:
19     print("Invalid input. Please enter an integer.")
20
```

Average time

0.002 s

2.00 ms

Maximum time

0.003 s

3.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

Debug

Expected output

Actual output

1

1

1 2

1 2

1 2 3

1 2 3

1 2 3 4

1 2 3 4

1 2 3 4 5

1 2 3 4 5

Terminal

Test cases

Thunderstorm w...

In effect

Search

ENG

IN

16:27

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